

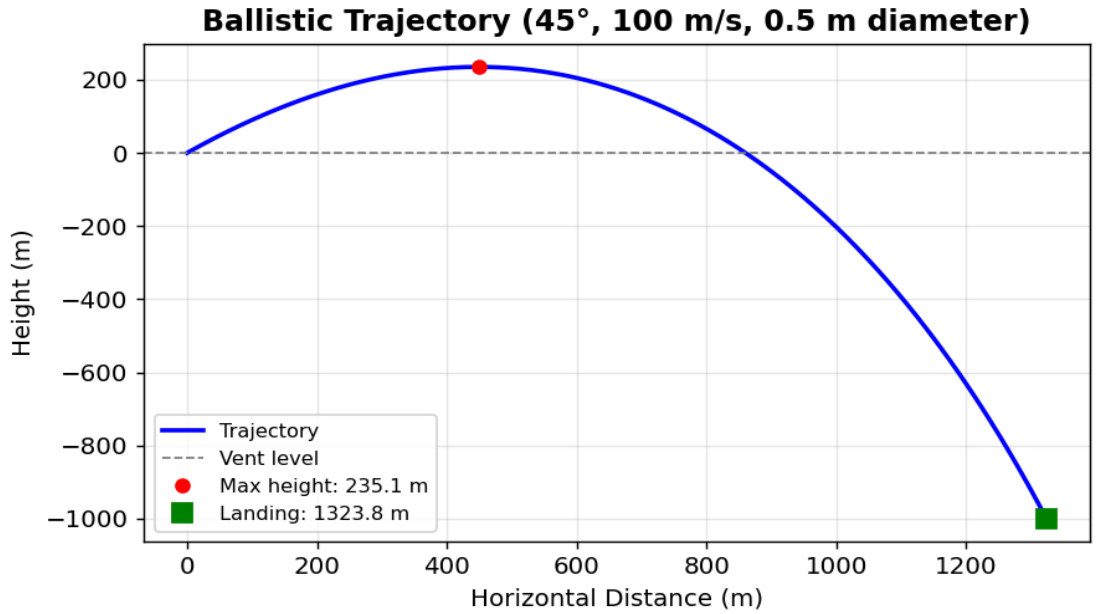
Photo by Richard Waitt, 1988 (U.S. Geological Survey).

Agenda

- What are ballistics?
- What is Eject!?
- Run Notebook 1: Eject! Basic and Notebook 2: Eject! With Topography in VICTOR to calculate flight paths
- Determine how far a block lands in 4 directions (N, E, S, and W)
- Produce a styled ballistic hazard layer in QGIS

What is Eject!?

- A calculator for **volcanic ballistic projectiles (VBPs)**, or fragments thrown from a volcano during an eruption
- It computes the **trajectory:** the curved path a rock follows through the air
- It tells you how far a block can land from the vent, and how high it flies
- It accounts for gravity, air drag, wind, and block size



One block: launched at 45°, 100 m/s, 0.5 m wide. The curve is its flight path; the green square is where it lands.

Hazard facts: Taddeucci et al. (2017); Brown et al. (2017).

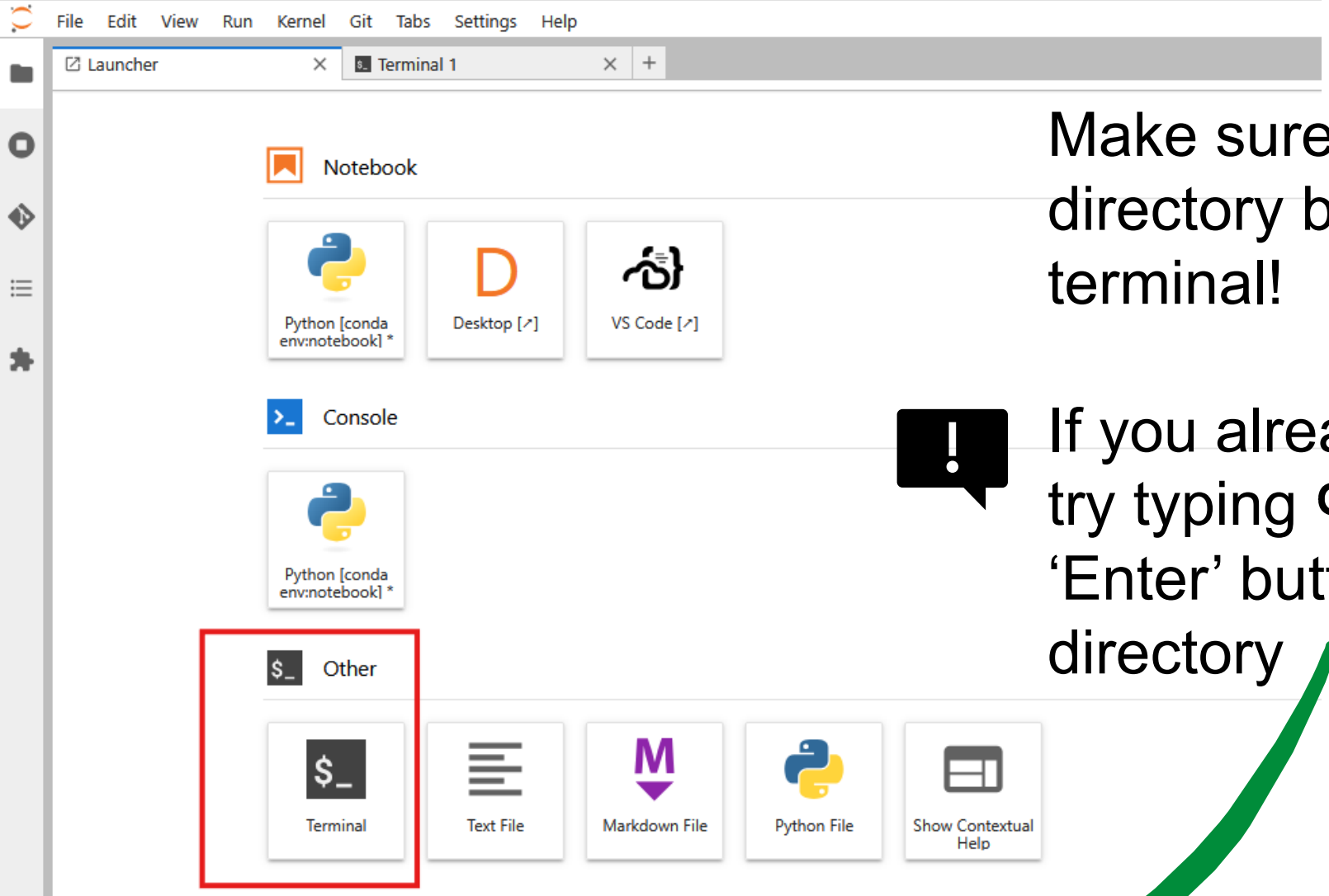
Table 4 Statistics for distance of fatal incidents by fatal cause, ordered by decreasing number of fatalities

Fatal Cause	Number of incidents	Number of fatalities	Min. distance (km)	Max. distance (km)	Mean distance (km)	Median distance (km)	2σ (km)
PDC	102	59,958	1.5	80	10.7	10.0	19.0
Tsunami	23	56,822	3.0	115	37.3	23.0	74.0
Lahars	72	49,938	1.0	100	24.1	20.0	38.9
Secondary lahars	41	6377	2.7	45	22.9	20.0	30.0
Tephra	52	4315	0.5	170	18.6	10.0	61.8
Avalanche	9	3525	1.0	20	8.3	7.0	13.2
Quiescent Gas	50	1698	0.1	74	10.9	7.5	30.8
Quiescent gas from vent	–	–	*0.0*	*13*	*5.4*	*6.8*	*9.0*
Lava flows	25	659	1.0	29	12.2	11.0	15.8
Lava flows from vent	–	–	*1.0*	*29*	*11.6*	*10.0*	*15.8*
Gas	16	585	5.0	21	9.8	9.0	13.1
Gas from vent	–	–	*1.0*	*21*	*7.2*	*5.0*	*15.1*
Ballistics	57	367	0.0	7	1.7	1.0	3.0
Hydrothermal	7	62	1.5	20	9.3	6.0	17.5
Hydrothermal from vent	–	–	0.8	3	1.6	1.3	1.7
Lightning	4	9	15.0	30	22.5	22.5	21.2
All eruptive hazards	515	214,004	0.0	170	13.5	8.6	35.6

Data include QL 1 and QL 2max. Gas, quiescent gas, lava flows and hydrothermal bursts are sometimes recorded with distance from their source vent as well as distance from the summit. In these cases we present all data under the parent category (e.g. quiescent gas) and substitute in the distance from vent data in the subcategory (e.g. quiescent gas from vent). We do not double count the fatalities and fatal incidents. All eruptive hazards includes the 'Multiple' and 'Eruption – unknown element' fatal cause categories: quiescent hazards are excluded. Note that the incident count includes sub-incidents



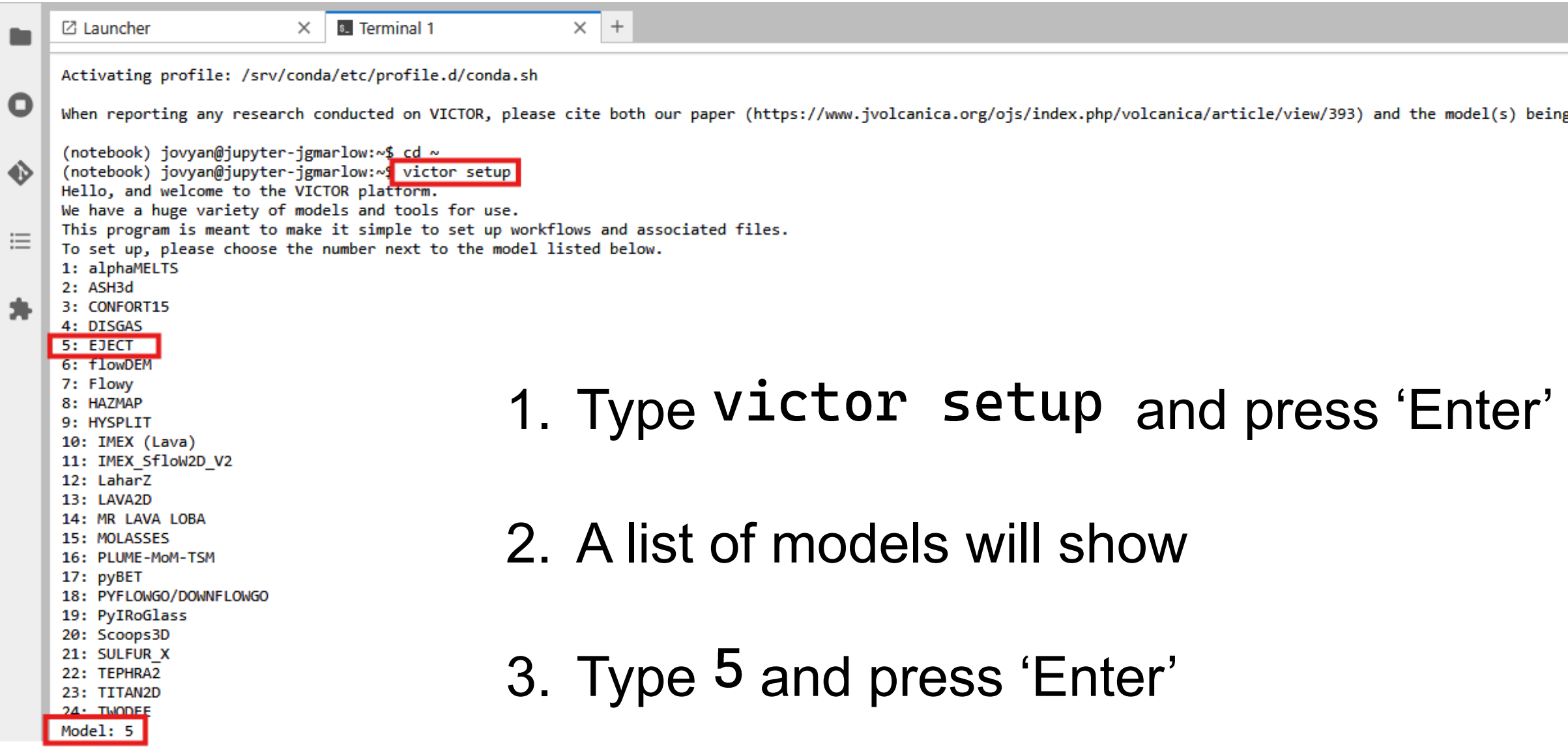
Figure 1. A roof impacted by ballistics during the 2000 eruption of Usu volcano (photo courtesy of Professor Okada).



Make sure to start in the home directory before launching the terminal!

If you already have a terminal open, try typing `cd ~` and pressing the 'Enter' button to change to your home directory





1. Type `victor setup` and press ‘Enter’
2. A list of models will show
3. Type `5` and press ‘Enter’

The model inputs

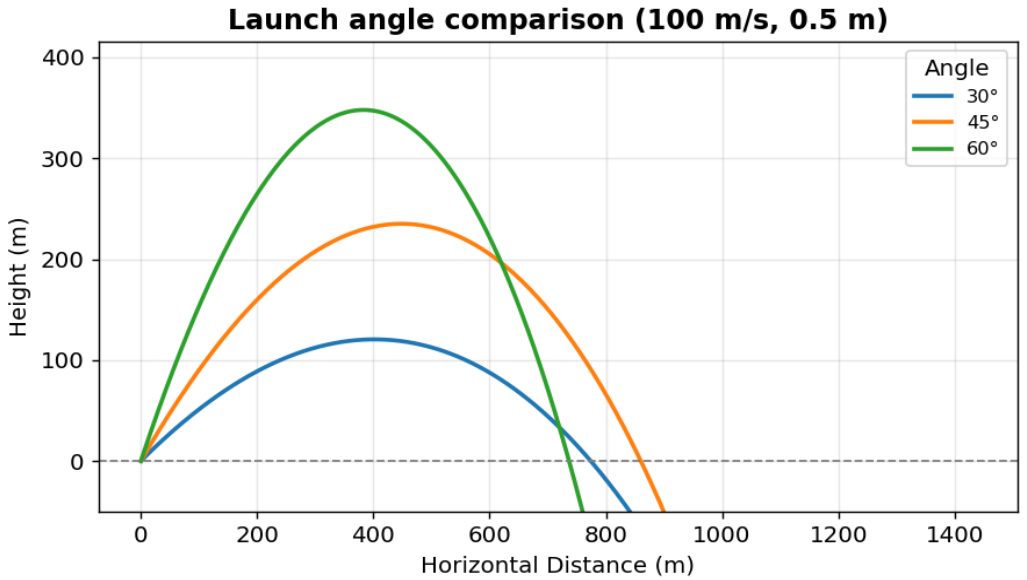
`eject(angle, velocity, diameter)`

Ejection angle	Initial velocity	Diameter
degrees (°)	meters per second (m/s)	meters (m)
How steep the launch is	How fast the block leaves the vent	How wide the block is
Low = flat & far	Faster = much farther	Bigger blocks are slowed less by air
High = steep & tall	**Typical values: 50–500 m/s**	**Typical values: 0.1–3 m**
Typical values: 15°–75°		

- We can change these inputs to model different eruptions
  - The eruption style sets these ranges: phreatic, Strombolian, and Vulcanian eruptions launch blocks at different speeds, angles, and sizes

Notebook 1: EJECT_basic.ipynb
Basic model: flat ground

- Models one block at a time and plots the trajectory
- Prints the max height and landing distance
- Change the inputs to compare angles, speeds, and block sizes side by side
- The landing distance is measured on a flat plane (z = -1000 m)



Comparing launch angles at 100 m/s. Lower angles travel farther; steeper angles fly higher but land closer.

How did changing the inputs impact the results?

- When you changed the **ejection angle**…?
  - With air drag, the farthest distance comes from an angle slightly below 45°
- When you changed the **velocity**…?
  - Biggest effect. Roughly double the speed → about 4× the range
- When you changed the **size**…?
  - Bigger blocks travel farther. They have more mass, so the air slows them less.

Part 1: The three inputs

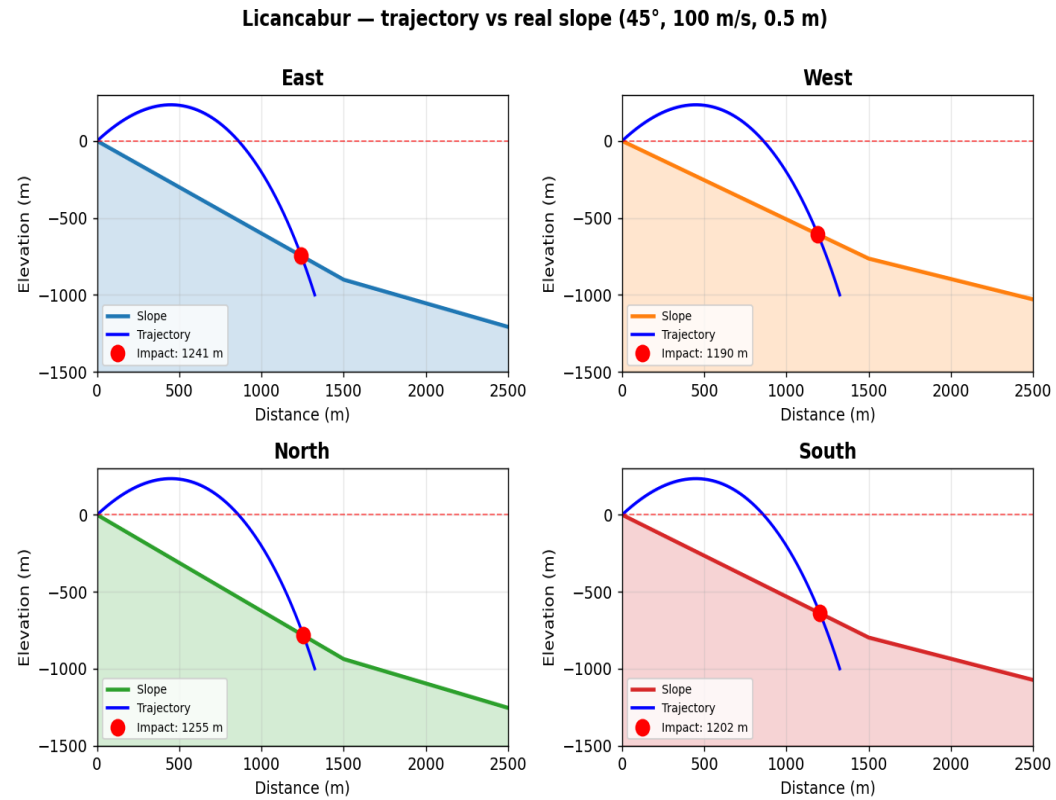
Every run uses `eject(angle, velocity, diameter)`:

Input	Units	Typical range	Meaning
ejection angle	degrees (°)	15–75	how steep the launch is
initial velocity	m/s	50–500	how fast the block leaves the vent
diameter	m	0.1–3	how wide (and heavy) the block is

Start with **45°, 100 m/s, 0.5 m** so everyone sees the same first result. Then change the numbers and try again.

Notebook 2: EJECT_withTopography.ipynb

- Real volcanoes are not flat!
- Eject! Calculates where the block actually hits the ground surface
- We will look at how far the block travels in 4 directions (N, S, E, and W)
- On the real slope, blocks land somewhat closer and a symmetric cone gives a near-circular zone.

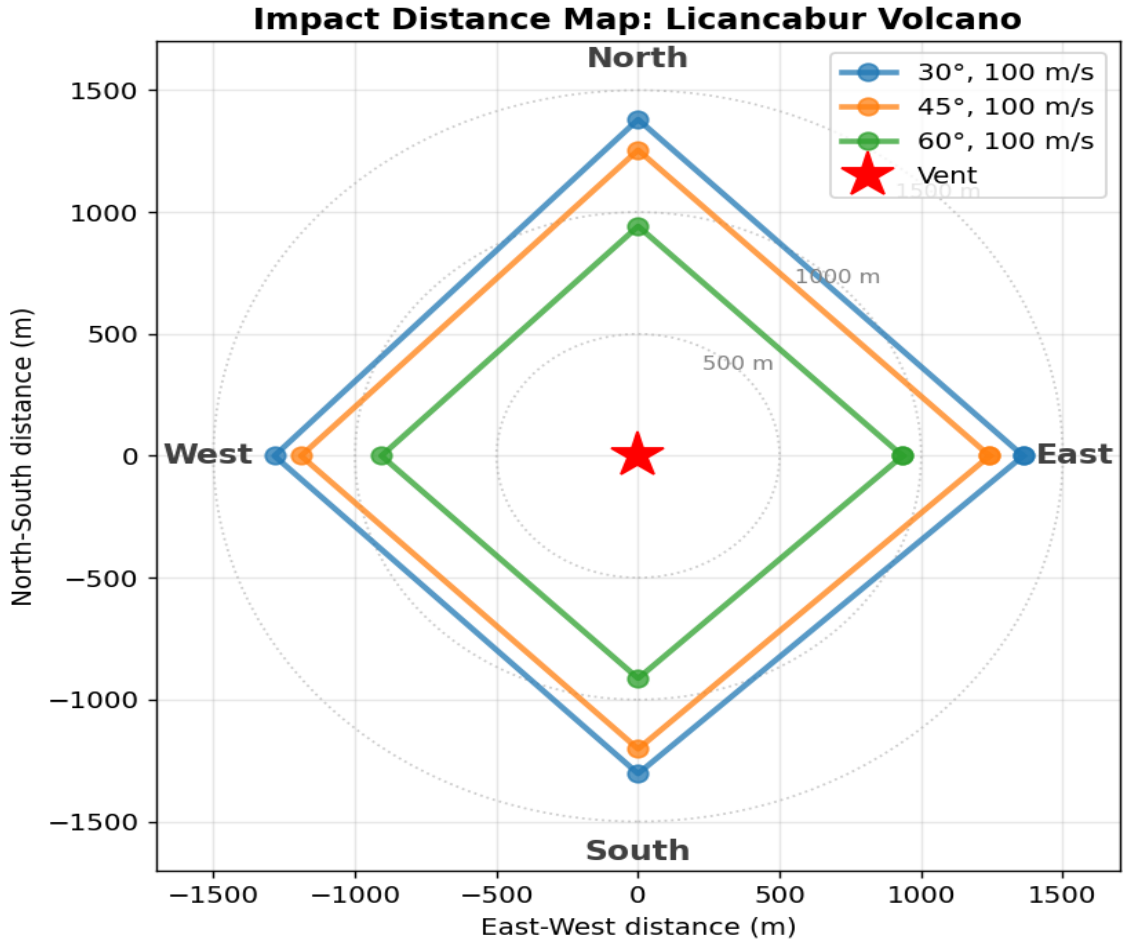


Same block, four directions. The blue arc is the trajectory; the shaded profile is the real slope; the red dot is the true impact.

Impact distances around the vent

- Notebook 2 prints a table of impact distances
- Below is the table for Licancabur

Direction	Impact distance
North	1,255 m
South	1,203 m
East	1,242 m
West	1,190 m



The real world is messier!!